

D1 Detailed Design

Project: HepatoFusion

Team members: Shivam Sinay Kharangate, Nipun Chandra

Partner: Cincinnati Children's Hospital

Document: Design D1 (Week 6). Sits beside D0_High_Level_Design. Does not overwrite D0.

AI use. An AI tool helped draft this document and the data-model figure. The data model, the outcomes join-and-clean algorithm, the file-schema contract, and the technology justifications are the team's decisions. We can explain them. The tool did not choose them.

Team decisions used in this document

Topic	Team decision
D1 scope	Detail Cohort case index (data model) and Outcomes cleaner (core algorithm).
Deferred to D2	Identifier gate, Radiology intake, Genomics intake, Modality integrator, Risk scorer, Review package (and PathPresenter wiring).
Store	JSON/CSV case files on approved shared folder. SQL (SQLite/Postgres) proposed later if joins get hard — not the D1 store.
Core algorithm	Outcomes join-and-clean only. Starting small while meeting modality leads. Risk/PHI gate deferred to D2.
API for now	No HTTP/integration API yet. Contract = versioned JSON/CSV schemas and validation rules.
Language	Python for these components.
Spreadsheet tooling	Reuse pandas + openpyxl in pipeline. Excel / Microsoft Graph optional for human-side workbook access.
Versioning	Versioned folder paths (v1/, v2/). Breaking field rename/remove requires a new folder version.

1. Header, scope, and conventions

Title: HepatoFusion

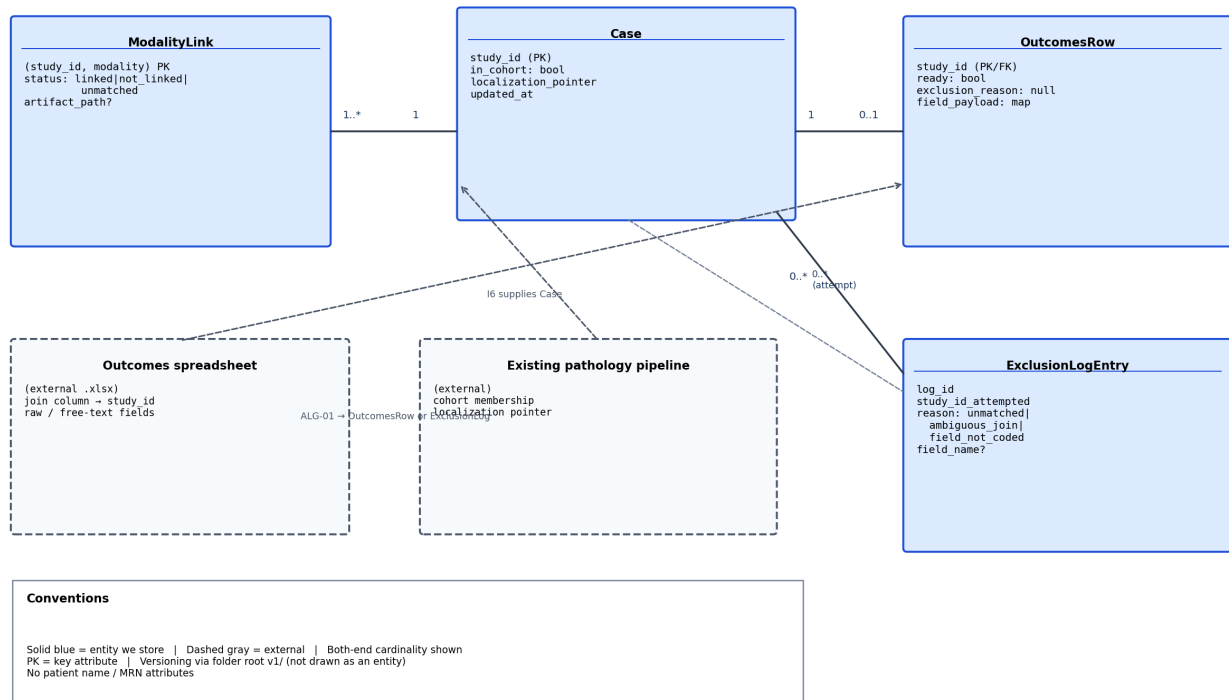
Goal (same wording as D0): Put an infrastructure in place for radiology, molecular data, and cleaned clinical outcomes on the existing pediatric liver-cancer cohort, then integrate those modalities with the finished pathology localization so a pathologist can use a research risk score as a helper. The finished pathology U-Net is also planned to surface inside PathPresenter (QuPath is the alternative viewer still under consideration), so the localization is available in the systems pathologists already use.

Scope. This D1 document details two D0 components: **Cohort case index** (holds the data model) and **Outcomes cleaner** (the computation that, if wrong, leaves the risk path with unmatched or uncoded rows). Identifier gate, Radiology intake, Genomics intake, Modality integrator, Risk scorer, and Review package are deferred to D2 while the team starts small and meets modality leads for radiology and molecular modeling.

Conventions for every figure

- A solid rectangle is an entity (or a component we build).
- An underlined attribute is a key (primary key or natural join key).
- A dashed rectangle is an external source or file outside the component.
- A line between entities is a relationship. Cardinality on both ends: 1, 0..1, or 1..*.
- Folder path labels such as v1/ mark the schema version, not a patient identifier.
- study_id is the internal cohort key. Patient name and medical record number are never attributes (US-05).

Editable figure source: D1_diagrams.drawio (sheet D1 Data Model).



2. Data model (the D1 diagram)

Component detailed: Cohort case index (D0). **Store choice:** simpler file store — one case folder per study_id under ../hepato_fusion/v1/cases/<study_id>/, holding JSON and CSV. Not a relational database in D1. A SQLite or Postgres migration is proposed later if multi-table joins become the bottleneck.

Entities and key attributes

Entity	Key attributes	Notes
Case	study_id (PK), in_cohort, localization_pointer, updated_at	One object per existing pathology-cohort case (102+ patients).
ModalityLink	(study_id, modality) PK; status ∈ {linked, not_linked, unmatched}; artifact_path?	modality ∈ {radiology, molecular, pathology}.
OutcomesRow	study_id (PK/FK), ready, exclusion_reason, field_payload	Only ready=true rows are model-ready (US-04, UC-02).
ExclusionLogEntry	log_id, study_id attempted, reason, field_name?	Audit of rejected rows. Study ID + field name, not patient name (I5).

Relationships and cardinality

Relationship	Cardinality	Why
Case → ModalityLink	1 → 1..*	Modality slots belong to a case; not many-to-many.
Case → OutcomesRow	1 → 0..1	At most one cleaned outcomes object per case; failed rows go to ExclusionLogEntry.
Case → ExclusionLogEntry	1 → 0..*	Many failed attempts can mention the same study ID.
Pathology pipeline → Case	external 1 → 0..1	I6 supplies membership + localization pointer.

Structural decisions

Decision	Choice	Why	Req / iface
Case is an entity, not only a folder name	Entity Case	Needs in_cohort and localization_pointer.	US-03, I6, I10
Modality status not booleans on Case	ModalityLink	Modalities arrive at different times; avoids schema churn.	US-02, US-03, I7, I8

Decision	Choice	Why	Req / Iface
Exclusion is not an OutcomesRow	ExclusionLogEntry	Failed join must not look scorable.	US-04, UC-02, I5, I9
Outcomes fields as payload map	field_payload	Field list still settling with leads; index remains study_id.	US-04
Case-Outcomes 1-0..1 not 1-*	One object per case	Risk path needs one outcomes slot.	UC-02, I9
Indexed / lookup field	study_id	Every join/open by study ID; fine at 102 and at 100× for files.	I5-I10
Store	JSON/CSV under v1/	Matches D0 file pipeline; no DB admin yet.	D0 D-01, D-03

3. Core algorithms

ALG-01 — Outcomes join-and-clean

What it is and the problem it solves. Takes the clinical-outcomes spreadsheet and the pathology cohort list, and emits either one model-ready outcomes object per matched case or an exclusion record. Solves US-04 / UC-02: the risk path must not train or score on free-text or unmatched patients.

Inputs and outputs (exact types).

	Type	Notes
Input source_workbook_path	str (path to .xlsx)	Approved storage only.
Input cohort_study_ids	set[str]	102+ study IDs (not names).
Input agreed_fields	list[str]	Must be non-empty and coded/numeric.
Input join_column	str	Spreadsheet column with study ID.
Output ready_rows	list[OutcomesRow] → outcomes.json	ready=true, exclusion_reason=null.
Output exclusions	list[ExclusionLogEntry] → JSONL	Under v1/logs/.
Output summary	dict[str, int]	{source_rows, kept, excluded}.

Expected complexity at realistic size. Let n = spreadsheet rows (hundreds to low thousands) and c = cohort size (~102). Build cohort hash set in $O(c)$, scan rows in $O(n \cdot k)$ with $k = |\text{agreed_fields}|$. Dominant cost is Excel I/O. At 100×, the join stays linear; wall time is still workbook read. Difference matters for operator wait on a huge sheet, not for asymptotic redesign — unless sheets become multi-hundred-MB, which would trigger the SQL migration discussion (§6).

Why this one rather than the alternatives.

Alternative	Why not now
Fuzzy name matching	Violates privacy and invents matches (UC-02 E1 forbids guessing).
Keep all rows; fix at train time	Pushes US-04 failure into the model; D0 blocks score when outcomes not ready.
Full SQL warehouse ETL	Premature while modality field lists are still settling; file cleaner matches D0.

Edge cases.

Case	Behavior
Empty workbook	kept=0, excluded=0, source_rows=0; no Case updates.
Duplicate study IDs in sheet	No last-writer-wins; both → exclusion ambiguous_join.
Join key matches 0 cases	Exclusion unmatched.
Join key matches >1 case	Exclusion ambiguous_join.
Matched but free-text/empty field	Exclusion field_not_coded + field_name (UC-02 A1).
Missing join column	Fail the run (E_MISSING_JOIN_COLUMN); no silent rename.

Pseudocode (optional).

```
cohort = set(cohort_study_ids)
for row in read_xlsx(source_workbook_path):
    key = row[join_column]
    if key not in cohort: exclude(unmatched); continue
    if not all_fields_ready(row, agreed_fields): exclude(field_not_coded); continue
    if key already emitted this run: exclude(ambiguous_join); continue
    write OutcomesRow(study_id=key, ready=true, field_payload=coded(row))
write summary counts
```

4. Build-versus-reuse decisions

Piece	Build/reuse	Library / service	License	Reason
Outcomes join-and-clean logic	Build	—	—	Project-specific rules (US-04 / UC-02).
Case folder + case_link.json	Build	—	—	HepatoFusion modality slots / D0 IDs.
Read .xlsx workbook	Reuse	pandas + openpyxl	BSD-3 / MIT	Mature; do not hand-roll Excel parsing.
Human workbook access	Reuse	Excel / Microsoft Graph	Microsoft terms	Editors only; check hospital allow-list.

Piece	Build/reuse	Library / service	License	Reason
JSON read/write	Reuse	Python json	PSF	Trivial, stable.
CSV export	Reuse	pandas / csv	BSD-3 / PSF	Standard tabular interchange.
DICOM / radiology features	Defer D2	pydicom (candidate)	MIT	Out of D1 scope.
Risk model training	Defer D2	sklearn / PyTorch	BSD-style	Not the D1 algorithm.
Auth / identity	Reuse / external	Hospital/UC folder access	Institutional	Do not hand-roll auth.
Dates / sorting if needed	Reuse	pandas	BSD-3	Do not hand-roll.

Maturity, licensing, performance, and fit were checked for pandas and openpyxl: widely used in research pipelines, permissive licenses, adequate for our workbook sizes, familiar from the pathology Python stack.

5. API contract

Status. There are no HTTP endpoints and no integration API yet. The contract for D1 is the versioned file schemas under `hepato_fusion/v1/` and the validation rules that decide whether a write is accepted. Callers are other pipeline stages (or a teammate's script) on approved storage.

Schema: `v1/cases/<study_id>/case_link.json`

Field	Type	Req	Valid range / notes
<code>study_id</code>	string	yes	Equals folder name; no patient name
<code>in_cohort</code>	boolean	yes	true for 102+ cohort members
<code>localization_pointer</code>	string null	yes	Relative path or null
<code>modalities</code>	object	yes	Keys $\subseteq$ radiology, molecular, pathology
<code>modalities.*.status</code>	string	yes	linked not_linked unmatched
<code>modalities.*.artifact_path</code>	string null	no	Set when linked
<code>outcomes.status</code>	string	yes	model_ready not_model_ready absent
<code>outcomes.path</code>	string null	no	Path to outcomes.json when ready

Schema: `v1/cases/<study_id>/outcomes.json`

Field	Type	Req	Valid range
<code>study_id</code>	string	yes	Same as case
<code>ready</code>	boolean	yes	Must be true in this file
<code>exclusion_reason</code>	null	yes	Must be null when ready
<code>field_payload</code>	object	yes	Keys $\subseteq$ agreed_fields; coded/numeric (not free text)

Schema: `v1/logs/outcomes_exclusions.jsonl`

Field	Type	Req	Valid range
<code>study_id_attempted</code>	string null	yes	May be null if join column empty
<code>reason</code>	string	yes	unmatched ambiguous_join field_not_coded
<code>field_name</code>	string null	no	Required when reason = field_not_coded

Logical operations (file-level methods)

Name	Inputs	Outputs	Errors
<code>validate_and_write_case_link</code>	path, JSON body	Written <code>case_link.json</code>	E_SCHEMA; E_ID_MISMATCH; E_PHI_FIELD
<code>run_outcomes_clean</code>	workbook, cohort list, agreed_fields, join_column	outcomes.json + exclusions JSONL + summary	E_IO; E_MISSING_JOIN_COLUMN; E_EMPTY_COHORT; E_AGREED_FIELDS_EMPTY
<code>read_case_record</code>	study_id	case_link + outcomes if present	E_NOT_FOUND; E_SCHEMA

Example request (into `run_outcomes_clean`)

```
{
  "source_workbook_path": "/approved/share/outcomes/source.xlsx",
  "cohort_list_path": "/approved/share/hepato_fusion/v1/cohort_study_ids.json",
  "join_column": "study_id",
  "agreed_fields": ["event_code", "followup_months", "vital_status_code"]
}
```

Example success summary response

```
{
  "schema_root": "hepato_fusion/v1",
  "source_rows": 120,
  "kept": 98,
  "excluded": 22,
}
```

```
"exclusion_breakdown": {"unmatched": 10, "ambiguous_join": 2, "field_not_coded": 10}
}
```

Example error response

```
{
  "ok": false,
  "error": "E_MISSING_JOIN_COLUMN",
  "message": "join_column study_id not present in workbook header",
  "study_id": null
}
```

Versioning. What is versioned is the folder schema root (v1/, later v2/). How: new directory tree for a new schema generation; old trees remain readable. A **breaking change** is renaming, removing, or changing the meaning of a required field in case_link.json or outcomes.json, or changing allowed status enums. Adding an optional field is non-breaking and may stay in v1/.

6. Technology choices with justification

File store (JSON/CSV under versioned folders). Skill fit: both teammates already work with files and Python dicts; no DB ops skill required this semester. Licensing: formats are open; no DB license. Community support: JSON/CSV are universal. Performance: fine at ~102 cases; one file open per study ID. Cost and hosting: existing approved CCHMC/UC shares — no new cloud bill; aligns with Week 3 privacy. **Alternative passed over:** Postgres/SQLite now — stronger joins, but adds admin and a service boundary D0 rejected for collection; kept as a proposed migration if n and join complexity grow.

Python for Outcomes cleaner and index writers. Skill fit: pathology U-Net work is already PyTorch/Python; data work is Python-heavy. Licensing: PSF for CPython; permissive scientific stack. Community support: dominant for research data cleaning. Performance: adequate for Excel-scale cleaning; not a real-time service. Cost and hosting: same approved compute as existing research jobs. **Alternative passed over:** Node.js/TypeScript — fine for web APIs we do not have yet; weaker fit to the ML/data stack.

pandas + openpyxl for spreadsheet I/O (pipeline). Skill fit: standard in the team's data work. Licensing: BSD-3 / MIT — compatible with academic research. Community support: very large; well documented Excel edge cases. Performance: enough for current workbook sizes (see ALG-01). Cost and hosting: free, local to approved machines. **Alternative considered:** Microsoft Excel / Graph for humans editing the sheet — reuse for stakeholders, but Graph is proprietary and network-dependent; hospital policy must allow it. Pipeline source of truth remains the file on the share.

No application front end in D1. PathPresenter remains the planned viewer (D0); D1 does not introduce a web UI. Skill / license / community / performance / cost: avoiding a custom UI keeps cost at zero and avoids a cloud client (rejected in D0). **Alternative passed over:** custom React viewer — duplicates PathPresenter and pulls toward deployment before Spring 2027.

Hosting / runtime. Approved institutional or hospital compute and shared folders only; nothing in the course Git repo that is PHI. Skill fit: matches current research practice. Licensing: institutional agreements already govern the cohort. Community support: N/A beyond hospital IT. Performance: GPU/CPU already used for the U-Net; cleaner is CPU/light. Cost: no new vendor. **Alternative passed over:** consumer cloud hosting — blocked by Week 3 / US-05 constraints.

7. Decision log (D1)

ID	Decision	Alternatives	Why
D1-01	Detail Cohort case index + Outcomes cleaner	Risk scorer first; all eight components	Unblocks collection; modality leads still shaping radiology/molecular.
D1-02	JSON/CSV file store now	SQLite/Postgres now	Matches D0; SQL proposed if joins/I/O hurt.
D1-03	Document ALG-01 only	Fully document risk + PHI gate	Start small; others to D2.
D1-04	File schemas as API contract	HTTP or Python RPC now	No integration API yet.
D1-05	Version with v1/ folders	schema_version field only; git tags only	Incompatible trees obvious on the share.
D1-06	pandas+openpyxl in pipeline; Excel/Graph optional	Graph as only I/O path	Pipeline must run on files without interactive Excel.

Traceability to D0 / stories

D1 piece	D0 component / interface	Stories / use cases
Case + ModalityLink	Cohort case index; I6–I10	US-02, US-03
OutcomesRow + ExclusionLog + ALG-01	Outcomes cleaner; I5, I9	US-04, UC-02
No PHI attributes	Identifier gate deferred; constraints	US-05
Risk / PathPresenter	Deferred D2	US-01, US-02